

DRL-Based Joint Trajectory-Frequency Design for UAV Anti-Jamming Communications

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Abstract—With the widespread applications of unmanned aerial vehicles (UAVs) in various fields, the security and anti-jamming capabilities of UAV communications have become key challenges. In this paper, we propose a method for joint optimization of UAV trajectory and frequency based on deep reinforcement learning (DRL), aiming to improve communication performance in dynamic environments containing intelligent reactive jamming. By constructing a Markov decision process (MDP) model incorporating spectrum and position states, we design a dual-branch deep Q-network (DQN) algorithm for joint trajectory-frequency decision-making. This enables the UAV to the dynamic adjustment of flight direction and communication frequency band to minimize trajectory deviation and evade jamming attacks. Finally, simulation results demonstrate that the proposed algorithm significantly improves communication performance against intelligent reactive jamming.

Keywords—UAV communications, trajectory control, channel selection, anti-jamming, deep reinforcement learning.

I. INTRODUCTION

The unmanned aerial vehicle (UAV)-assisted communication systems offer advantages such as high flexibility and stable coverage. Trajectory control is the prerequisite and key for UAVs to accomplish various predefined tasks. For example, in [1], a multi-objective path planning framework was proposed, enabling UAVs to explore paths in dynamic urban environments safely. Authors in [2] employed the A* algorithm for two-dimensional path planning of multi-rotor UAVs to analyze high-altitude safe trajectories, ensuring safe flight and obstacle avoidance. In [3], a precise three-dimensional path planning method for UAVs based on an enhanced multi-objective swarm intelligence algorithm is designed, optimizing objectives based on total path length and terrain threat level. However, due to the dominant Line-of-Sight (LoS) nature of air-to-ground channels, UAV communications are highly vulnerable to malicious jamming. With the integration of software defined radio and communication countermeasures, intelligent jammers possessing sensing capabilities and multiple jamming patterns pose a serious threat to UAV communication security. Therefore, trajectory control remains one of the key issues in anti-jamming communication for UAVs. In [4], a path planning algorithm for UAV anti-jamming communication was proposed, utilizing signal power maps and signal-to-noise ratio (SNR) maps for UAV path planning. In [5], the UAV's trajectory was

optimized using block coordinate descent and successive convex approximation methods to enhance the communication rate under jamming attacks. In [6], system throughput was improved by jointly optimizing the UAV's trajectory and communication time slots. In [7], an efficient iterative algorithm was proposed for UAV trajectory design to maximize the minimum expected value among ground sources (GS) within a limited flight period. However, in practical the jammer may dynamically employ jamming patterns to improve its jamming effect. This makes it difficult for UAVs to acquire accurate jamming information, potentially degrading anti-jamming performance.

To counter complex multi-pattern jamming, numerous existing studies employ reinforcement learning (RL) methods to learn jamming patterns and make intelligent anti-jamming decisions. For example, in [8], a model-free RL algorithm was innovatively introduced, utilizing the Q-learning (QL) algorithm to derive dynamic anti-jamming spectrum access strategies [9]. In [10], a deep Q-network (DQN) framework was adopted to improve the channel selection process, demonstrating superior cumulative reward performance compared to traditional QL algorithms. The authors of [11] and [12] also proposed anti-jamming communication methods from a DRL perspective to obtain optimal anti-jamming policies. However, most current research primarily focuses on using DRL to solve the UAV's spectrum access decision problem, where the state complexity and decision space are relatively small. For UAV communication against intelligent jamming, joint optimization with the flight trajectory is required. However, introducing trajectory optimization will greatly increase the anti-jamming optimization problem.

To address the above challenges, this paper considers a joint trajectory and channel optimization problem for countering intelligent reactive jamming. The jammer continuously senses communication signals, and releases sweeping jamming when the signal strength falls below a threshold. If the signal exceeds the threshold, it releases tracking jamming for precise attacks. The UAV counters the jammer's signal sensing and attacks through trajectory control and channel selection. The contributions of this paper are summarized as follows: 1) We investigate a practical UAV communication system, considering the selection of the UAV trajectory and communication channels. 2) We design a DRL-based algorithm for joint trajectory and channel optimization of the UAV anti-jamming communication. We decouple the joint trajectory-channel decision-making, designing separate

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reward functions for each decision using a parallel learning network framework. 3) Simulation results demonstrate that the proposed method effectively evades tracking jamming and sweeping jamming.

The subsequent structure of this paper is arranged as follows: In Section II, the system model is introduced, detailing the system model, communication model, and intelligent reactive jamming model, while presenting the problem formulation. In Section III, a joint trajectory-frequency optimization algorithm based on parallel deep reinforcement learning is designed. In Section IV, simulation results are provided. Finally, Section V concludes the paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

Fig. 1 shows the UAV communication scenario, which includes multiple ground users ($\mathcal{M} = \{1, 2, \dots, M\}$), a UAV, and a ground jammer. The source and destination positions of the UAV are fixed, with preset trajectory and flight direction. The key concept is for the UAV to deviate from its original trajectory as little as possible without being attacked by the jammer. UAV schedules users with the best channel conditions and communicates during flight. The jammer has sensing capability and multiple jamming patterns to attack communication. We assume that the user's position remains constant, and the UAV's position is considered approximately unchanged within a given time slot. We assume the height of the UAV is fixed, then the coordinate of the UAV in the t -th time slot is $q_t = [x_t, y_t, H]$, and user m 's position is fixed at $q_m = [x_m, y_m, 0]$. The fixed jammer's coordinate is $q_j = [x_j, y_j, 0]$.

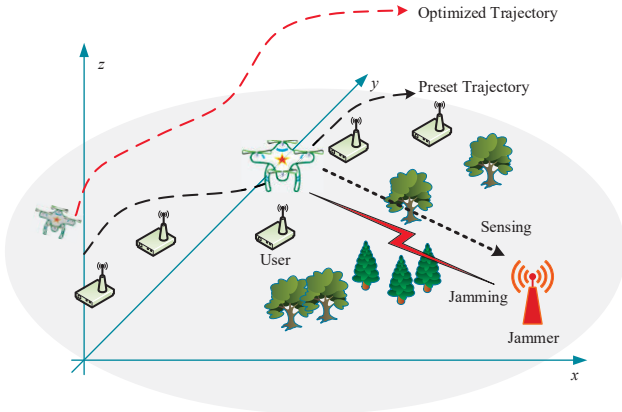


Fig. 1. System Model.

Users can transmit data according to time slots and achieve dynamic spectrum anti-jamming communication through channel selection. Fig. 2 shows the working time slot diagram of the UAV and the jammer. Among them, the UAV time slot consists of three parts: 1) Sensing stage: The UAV senses environmental spectrum states and processes them as inputs for the intelligent learning algorithm; 2) Transmission stage: The UAV selects the transmission channel based on the algorithm, and communicates with the user in the time-division duplex mode; 3) Learning stage: The UAV updates the learning algorithm based on sensing data and communication results.

In this paper, the time division multiple access (TDMA) strategy is utilized when the UAV communicates with the

users. In each time slot, the UAV communicates with only one user through scheduling. Let $\alpha_{m,t}$ represents the binary user scheduling variable, where $\alpha_{m,t} = 1$ denotes a communication connection being established between the UAV and the m -th user device in period t , otherwise $\alpha_{m,t} = 0$.

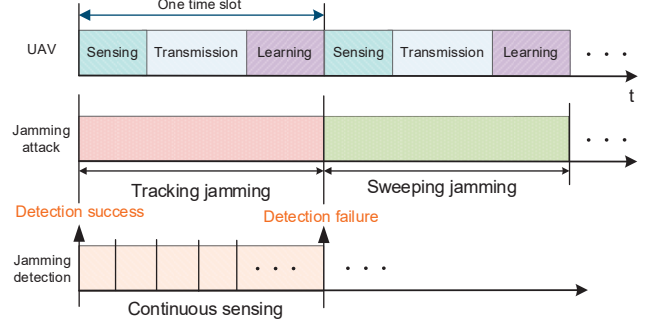


Fig. 2. Time slot diagram of UAV and jammer.

B. Communication Model

The UAV communication frequency band $[f_L, f_U]$ is uniformly divided into K non-overlapping channels, and the set of center frequencies is represented as $\mathcal{F} = \{f_1, f_2, \dots, f_K\}$. Considering the line-of-sight-dominant characteristic of air-to-ground channel, we use the free space propagation to model the LoS channel, which is represented as $h = \beta_0 d^{-\xi}$. Where β_0 is the reference path loss constant, and ξ is the channel attenuation exponent factor. d is the distance between the UAV and the user.

The Signal-to-Interference-plus-Noise Ratio (SINR) of the received signal by the user is calculated as follows:

$$\beta_t = \frac{p_t h_{t,r}}{p_j h_{j,r} \delta(f_j^t = f_u^t) + \sigma^2} \quad (1)$$

where p_t is the power of the user, p_j is the transmission power of the jammer. f_j^t and f_u^t represent the jammer frequency and the user center frequency at t , respectively. $\sigma^2 = \int_{f_L - b_u/2}^{f_U + b_u/2} n(f) df$ denotes the noise power. $n(f)$ represents the power spectral density (PSD) function of the background noise and $b_u = (f_U - f_L)/K$ represents the bandwidth. $h_{t,r}$ and $h_{j,r}$ respectively represent the channel gain from the UAV transmitter to the receiver and the channel gain from the jammer to the receiver. $\delta(\cdot)$ represents the indicator function, which is 1 when the condition is met and 0 otherwise. Based on the received SINR, the communication rate $V_{m,k}$ of user m on channel k can be calculated, where $V_{m,k}$ is

$$V_{m,k} = \begin{cases} b_u \cdot \log_2(1 + \beta_t) & \beta_t \geq \beta_{th} \\ 0 & \beta_t < \beta_{th} \end{cases} \quad (2)$$

where β_{th} represents the signal-to-noise ratio threshold for successful demodulation by the receiver. When the received SINR is below the demodulation threshold, the user is interfered and the communication rate is 0.

To better understand the spectrum environment, we introduce the generation principle of spectrum sensing data based on the PSD function. Considering all signals coexisting simultaneously, the PSD function of the signal at the receiver is

$$s(f, t) = h_{t,r} U(f - f_u) + h_{j,r} U_j(f - f_j) + n(f) \quad (3)$$

where $U(f - f_u)$ and $U_j(f - f_j)$ represent the PSD of the useful signal and the jamming signal.

The definition of discrete spectrum sampling values is

$$o_i = 10 \log_{10} \left\{ \int_{i\Delta f}^{(i+1)\Delta f} s(f, t) df \right\} \quad (4)$$

where Δf is the resolution of spectrum analysis, and the result obtained from one spectrum sensing is $O_t = [o_1, o_2, \dots, o_X]$, where X is the number of sampling points.

C. Jamming Model

We consider an intelligent reactive jammer which can monitor communication signals and release multiple jamming patterns according to the states of communication signals. Assuming that the jammer has strong response capability and the sensing response time is so short that it is negligible. When the spectrum monitor of the jammer detects the communication signal, it immediately releases the tracking jamming with the same frequency of communication signal. When no communication signal is detected, it will release the indiscriminate jamming such as sweeping jamming. Assuming that the jammer detects signals based on energy detection [13], the detection threshold is set to α_{th} , and the power of the UAV signal received by the jammer is

$$\alpha_t = h_{t,j} \int_{f_u^t - b_u/2}^{f_u^t + b_u/2} U(f, t) df + \sigma^2 \quad (5)$$

where $U(f, t)$ represents the PSD function of the UAV's baseband signal and σ^2 is the noise power. When the jammer detects that the received energy of the signal α_t is greater than the threshold α_{th} , the communication signal is successfully detected, otherwise, the detection fails. To achieve concealment, the UAV can avoid tracking jamming by changing their flight trajectory.

D. Problem Formulation

In this paper, our goal is to maximize the communication rate when the UAV communicates with the user by jointly optimizing the trajectory control and channel selection of the UAV. This means minimizing the UAV's flight path while successfully avoiding tracking jamming and sweeping jamming, while improving the communication rate between the UAV and the user. Therefore, the optimization problem can be expressed as follows:

$$\max_{\{f_k, q_t\}} \frac{1}{T} \sum_{t=1}^T \sum_{m=1}^M V_{m,k} \quad (6)$$

$$\text{s.t.} \quad C1: q_{t+1} = q_t + \vec{V} * \sigma \quad (6a)$$

$$C2: 0 \leq x_n \leq x_{\max}, 0 \leq y_n \leq y_{\max} \quad (6b)$$

$$C3: \alpha_{m,n} \in \{0, 1\}, \forall m \in M \quad (6c)$$

$$C4: \sum_{m=1}^M \alpha_{m,n} \leq 1, \forall m \in M, \forall n \in N \quad (6d)$$

$$C5: f_k \in \{f_1, f_2, \dots, f_K\} \quad (6e)$$

where constraint (C1) denotes the UAV flying constraint and $\vec{V}$ is the speed of the UAV. Meanwhile, constraint (C2) ensures that the range of UAV flight is within the designated area. Constraints (C3) and (C4) represent communication

scheduling constraints for user devices. Finally, (C5) represents the selected channel.

In fact, the problem in (6) is difficult to solve because it is a joint optimization problem involving optimization of UAV trajectory and selection of channels. To address this issue, we introduced a DRL method to intelligently design the trajectory and channel of the UAV, while considering changes in jamming patterns.

III. PROPOSED SOLUTION

In this section, we first transform the established problem into a Markov Decision Process (MDP), and then use the DRL algorithm to optimize the trajectory of the UAV and select the access channel to maximize the communication rate between the UAV and users.

A. MDP Formulation

Based on the established optimization problem, we transform the original optimization problem into an MDP architecture and use the DRL algorithm to solve it. The four key elements of MDP are defined as follows.

1) *State s_t* : The UAV anti-jamming MDP inputs the spectrum waterfall diagram s_1 and UAV's position s_2 as state inputs. The spectrum waterfall diagram s_1 is constructed by concatenating historical instantaneous spectrum sensing results O_t :

$$s_1 = \begin{bmatrix} o_1^t & o_2^t & \dots & o_X^t \\ o_1^{t-1} & o_2^{t-1} & \dots & o_X^{t-1} \\ \vdots & \vdots & \vdots & \vdots \\ o_1^{t-\Phi+1} & o_2^{t-\Phi+1} & \dots & o_X^{t-\Phi+1} \end{bmatrix} \quad (7)$$

where Φ denotes the historical look-back length. Based on the above definition, the spectrum waterfall diagram contains three-dimensional information (time, frequency, power). The spectrum waterfall diagram not only reflects the dynamic changes of intelligent reactive jamming in the environment but also satisfies the Markov property, that is, the current state is only related to the previous state and action. And the marking of the UAV's position $s_2 = (x_t, y_t, H)$ can also better learn the optimal trajectory.

2) *Action a_t* : The UAV uses trajectory control and channel selection joint decision-making for covert anti-jamming communication, where trajectory control is used to evade tracking jamming detection, and channel actions are used to access idle channels. In this paper, we assume the UAV has a preset trajectory and will not fly in the direction opposite to the preset trajectory. Therefore, we define five flight actions for the UAV. Assuming the UAV is currently flying to the right, the flight direction action $a_t^d \in \mathcal{D}$, where $\mathcal{D} = \{0, 1, 2, 3, 4\}$ represents: upward, upper-right, straight forward, lower-right, and downward. The UAV's channel selection action is $a_t^f \in \mathcal{K}$, where $\mathcal{K} = \{1, 2, \dots, K\}$ is the set of selectable channels.

3) *Reward r_t* : The reward function is used to make the UAV learn the correct decision-making policy. During the dynamic communication between the UAV and the user, every step of the UAV's movement will receive a reward. In order to learn the optimal trajectory and channel selection, we designed two reward functions. The reward function of the channel decision is defined as

$$r_t^f = \begin{cases} \eta, & \text{No jamming} \\ 0, & \text{Tracking jamming} \\ -\eta, & \text{Sweeping jamming} \end{cases} \quad (8)$$

If the communication between the UAV and the user is not be interfered with, a position reward η is obtained. If the UAV is attacked by tracking jamming, the channel decision is meaningless, so a conservative reward of 0 is set. If jammed by sweeping jamming, the channel selection is incorrect and a negative reward $-\eta$ is obtained. The reward function of the trajectory decision network is defined as

$$r_t^d = \begin{cases} \zeta, & \text{No tracking jamming, no deviation} \\ -\mu \cdot d_t, & \text{No tracking jamming, deviation} \\ -\zeta, & \text{Tracking jamming} \end{cases} \quad (9)$$

where $\mu \in (0,1]$ is a constant coefficient of trajectory deviation penalty. When the UAV is on a preset trajectory and not being disturbed by tracking jamming, it receives a positive reward ζ . When it is not on the preset trajectory and not being disturbed by tracking jamming, it receives a negative reward proportional to the deviation distance $-\mu \cdot d_t$. When the UAV is detected to be tracked by tracking jamming, a negative reward $-\zeta$ is obtained to direct the UAV flying away from the jammer.

4) *State Transition Model*: The state transition model represents the way in which the UAV chooses to transition from the current state to the next state. The state transition model is not only determined by (6a), but also depends on the jamming mode used by the jammer.

B. Proposed Algorithm

In response to the characteristics of intelligent reactive jamming, this part designs a joint anti-jamming algorithm based on parallel DRL to solve the above-mentioned anti-jamming MDP. The proposed algorithm framework is shown in Fig. 3. We decompose the single decision network in traditional DQN into two parallel decision networks, decoupling the joint decision into two independent decisions. Spectrum sensing acquires the received power of each channel (including jamming signals and environmental noise) by monitoring the wireless environment in real-time, constructing a two-dimensional spectrum waterfall diagram. This waterfall plot serves as input features for the frequency decision network. After spatial feature extraction through convolutional layers, the features are passed to the output state value $Q(s,a)$ via dense layers. Meanwhile, the direction decision network takes positional states as input and outputs the optimal direction selection strategy after processing through dense layers. After each network outputs its corresponding decision, their respective network parameters are updated based on their reward value functions. The design of parallel decision networks mainly includes the following two reasons:

- Adapting to the dynamic characteristics of intelligent reactive jamming. Intelligent reactive jamming is dynamic and adversarial, simultaneously affecting the UAV's trajectory control and channel selection. The dual-branch structure of the parallel decision-making network allows for real-time adjustments to trajectory and channel selection: one branch dynamically plans flight directions based on the UAV's current position

and jamming pattern to evade tracking jamming, while the other branch rapidly switches communication channels based on channel conditions and jamming characteristics to maintain successful communication. This decoupled decision-making mechanism enables the UAV to respond more flexibly to complex jamming scenarios, thereby enhancing overall anti-jamming capability and communication success rate.

- Reducing the exploration difficulty of high-dimensional action space. Compared to traditional single-domain anti-jamming decision-making, this paper employs joint trajectory-channel decision-making, which significantly expands the action space and increases exploration difficulty. However, in environments with intelligent reactive jamming, certain action combinations exhibit redundancy. For example, channel selection becomes ineffective when trajectory decisions are incorrect. A single network would need to explore all possible action combinations ($10 \text{ channels} \times 5 \text{ directions} = 50$), whereas the parallel network only requires independent exploration of channels and trajectories ($10 \text{ channels} + 5 \text{ directions} = 15$) without compromising performance. This reduces computational complexity from multiplicative to additive scaling, enabling more efficient learning of the optimal policy.

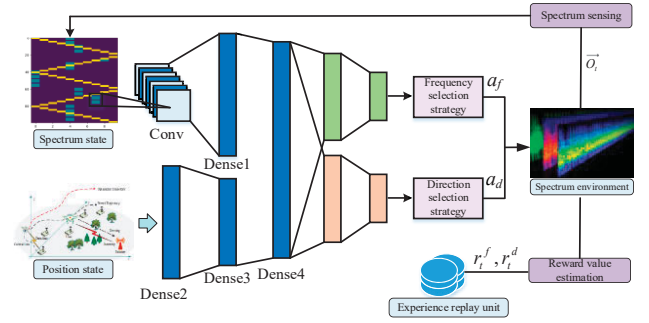


Fig. 3. Framework of the proposed algorithm.

Traditional DQN algorithm uses convolutional neural network to fit value function.

$$Q(s,a) = f(s,a;\theta) \quad (10)$$

where, function f represents the function fitted by the neural network, and θ is a parameter of function f , which is the weight of the network. The training process of the network updates the parameters. Therefore, the optimal Q value is represented as

$$Q(s,a;\theta) = \mathbb{E}[r + \max_{a'} Q(s',a';\theta) | s,a] \quad (11)$$

Using a quadratic cost function to calculate errors.

$$L(\theta) = \mathbb{E}\{[r + \gamma \max_{a'} Q(s',a';\theta) - Q(s,a;\theta)]^2\} \quad (12)$$

where $\max_{a'} Q(s',a';\theta)$ represents the value of taking the optimal action in the next state.

The pseudocode of the algorithm proposed in this paper is shown in Algorithm 1. Unlike the traditional DQN algorithm, parallel network parameters are updated based on joint loss values, calculated as

$$L_t^{f+d}(\theta_t) = \mathbb{E}_e[L_t^f(\theta_t) + cL_t^d(\theta_t)] \quad (13)$$

$$L_t^f(\theta_t) = \mathbb{E}_e \left[(y_f - Q_f(s_t, f_t; \theta_t))^2 \right] \quad (14)$$

$$L_t^d(\theta_t) = \mathbb{E}_e \left[(y_d - Q_d(s_t, d_t; \theta_t))^2 \right] \quad (15)$$

where $L_t^f(\theta_t)$ and $L_t^d(\theta_t)$ respectively represent the estimation errors of the channel decision network and the trajectory decision network, and $c \in (0, 1]$ is a constant coefficient. After calculating the loss function, update the network parameters using $\theta_{t+1} = \theta_t - \beta \nabla_{\theta_t} L_t^{f+d}(\theta_t)$.

In addition, to increase action exploration, this article uses the dynamic ϵ -greedy method, where the exploration factor dynamically decays with the number of iterations, calculated as

$$\epsilon_t = \begin{cases} \epsilon_0 e^{-\nu t}, & \epsilon \geq \epsilon_f \\ \epsilon_f, & \epsilon < \epsilon_f \end{cases} \quad (16)$$

where ϵ_0 is the initial exploration factor, ϵ_f is the termination exploration factor, and ν represents the fading coefficient. The user selects a random action with probability ϵ_t and a greedy action $a = \arg \max Q(s, a; \theta)$ with probability $1 - \epsilon_t$ when making a decision.

Algorithm 1 Joint Trajectory-Frequency Optimization Algorithm

- 1: **Initialize:** randomly generate estimated Q-value network parameters θ , target Q-value network that satisfies $\theta^- = \theta$, experience replay unit $\mathcal{C} = \emptyset$, and training sample $\mathcal{B}$;
 - 2: **for** episode = 1, 2, ..., L **do**
 - 3: The current state of UAV is $s_t = (s_1, s_2)$;
 - 4: Calculate the Q-value and select d_t and f_t based on the dynamic ϵ -greedy strategy;
 - 5: The UAV transmits data based on $a_t = (f_t, d_t)$;
 - 6: If disturbed:
 - 7: **for** m = 1, 2, ..., T **do**
 - 8: Maintain d_t unchanged, sense channel access, and record whether there is jamming;
 - 9: **end for**
 - 10: If all T detection time slots are jammed, the UAV is attacked by tracking jamming;
 - 11: Obtain r_t^f and r_t^d according to (8) and (9);
 - 12: Obtain the next state s_{t+1} ;
 - 13: Add $e_t = (s_t, f_t, r_t^f, r_t^d, s_{t+1})$ to $\mathcal{C}$;
 - 14: If $|\mathcal{C}| > |\mathcal{B}|$:
 - 15: Randomly sample training sample $\mathcal{B}$ from $\mathcal{C}$;
 - 16: Calculate the estimation error $L_t^{f+d}(\theta)$ according to equation (13);
 - 17: Using $\theta_{t+1} = \theta_t - \beta \nabla_{\theta_{t+1}} L_t^{f+d}(\theta_t)$ to update network parameters;
 - 18: Update the $\theta^- = \theta$ every N_t steps.
 - 19: **end for**
-

IV. NUMERICAL RESULTS

A. Simulation Settings

The ground users are positioned at (200m, 0m, 0m) and (300m, 0m, 0m), while the UAV's initial position is (0m, 200m, 50m) and its destination is (500m, 200m, 50m). The preset flight path follows the straight line $y=200m$. The jammer is located at (250m, 300m, 0m). The UAV repeatedly moves from the starting position to the ending position at a speed of $v = 20$ m/s. The communication time slot duration is set to 0.1 s. The available communication bandwidth is 20 MHz, evenly divided into 10 non-overlapping channels, each with a bandwidth of 2 MHz. The sensing module has a

frequency sampling interval of 0.1 MHz, with 200 sampling points per time. A full-band spectrum sweep takes 10 ms. The spectrum waterfall diagram retains a history length of 100. The user transmission power is 20 dBm/channel, the receiver SNR threshold β_{th} is 0 dB, and the background noise power is -80 dBm. When the jammer deploys sweeping jamming, the sweeping speed is 1 channel/step, with a jamming bandwidth of 2 MHz and power of 20 dBm. The signal detection threshold is -60 dBm. If detection fails, sweeping jamming continues; upon successful detection, the jammer switches to tracking jamming. The summary of hyperparameters for network training is as follows: discount factor is $\gamma = 0.9$, learning rate is $\alpha = 0.6$, training batch size is Batch_Size=32, memory replay unit size is 5000, initial exploration factor is $\epsilon_0 = 1$, termination exploration factor is $\epsilon_f = 0.01$, and the fading coefficient is 0.995. The proposed algorithm is implemented using TensorFlow 2.6 on hardware featuring an NVIDIA RTX 3060 graphics card, an Intel i7-11800H core processor, and 32GB DDR4 RAM.

B. Results and Analysis

Fig. 4 demonstrates the convergence of the proposed algorithm, presenting spectrum waterfall diagrams during both the initial and converged learning phases. Sweeping jamming is released when jammer detection fails while tracking jamming is deployed upon successful detection. The results reveal that during the initial learning phase, inappropriate UAV flight directions lead to smart reactive jamming detection in most time slots, followed by tracking jamming attacks. In the converged learning phase, all user signals successfully avoid overlapping with jamming signals, evading detection, and accessing idle channels. Consequently, the smart reactive jammer continuously generates sweeping jamming. Thus, the proposed algorithm effectively captures jamming dynamics in unknown dynamic environments, adaptively adjusting flight directions and channels to proactively evade jamming.

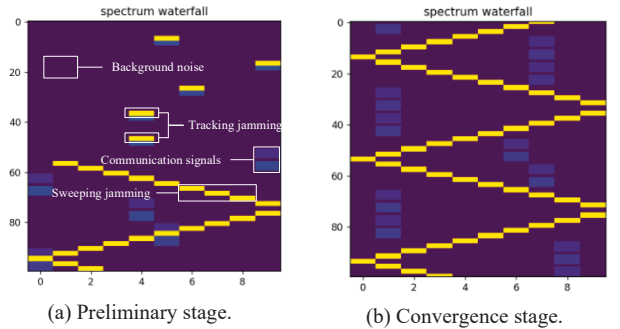


Fig. 4. Spectrum waterfall plot at different stages.

Fig. 5 presents the normalized reward values for channel selection and trajectory control obtained during UAV-user communication. The horizontal axis represents episodes, with the red curve indicating trajectory rewards and the blue curve representing channel rewards. The two curves show an overall upward trend, with a faster rate of increase in the early stages of training and gradually stabilizing in the later stages. We observe higher values in the red curve because trajectory rewards operate independently during training and are unaffected by channel selection. Conversely, channel rewards are influenced by flight direction, and the UAV selects channels vary at different positions. Consequently, channel rewards only begin to rise steadily in later stages after the trajectory is largely established.

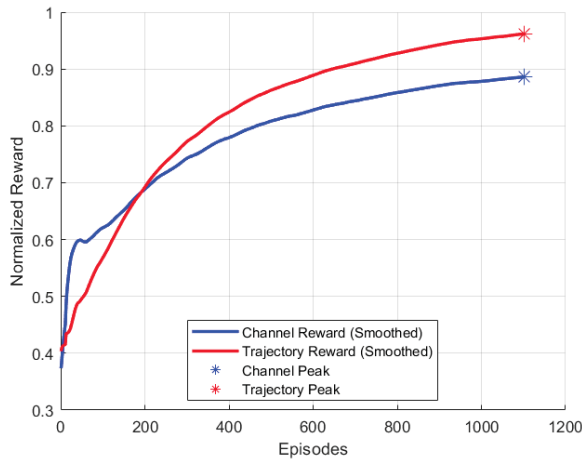


Fig. 5. Normalized reward values.

Fig. 6. presents a comparative diagram of communication rates achieved by the UAV and users under different schemes. We primarily consider two comparative methods: only trajectory control and only channel selection. The horizontal axis represents different scheme types, while the vertical axis indicates the final average communication rate obtained after 1000 training episodes for each scheme. As shown in the figure, the communication rate of the joint channel-trajectory approach proposed in this paper significantly exceeds those of the other two methods. This further demonstrates that our method successfully learns the behavioral patterns of intelligent reactive jammers and can adaptively adjust both trajectory and channel selection to enhance transmission rates.

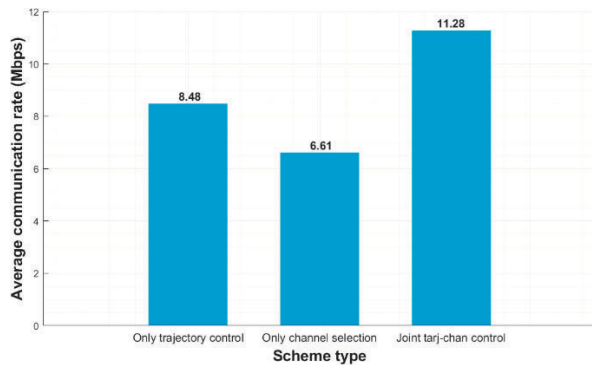


Fig. 6. Average communication rate under different schemes.

V. CONCLUSION

This paper concluded the challenge of covert communication for UAVs in dynamic jamming environments by proposing a DRL-based joint trajectory-spectrum optimization method. By constructing an MDP model for

spectrum state and positional state, we design a dual-branch DQN network that decouples trajectory direction from channel selection, utilizing independent reward functions to guide action learning separately. The experimental results show that the algorithm dynamically adjusts the trajectory and channel. During the jamming detection stage, it changes the trajectory to evade tracking attacks while accessing idle channels to avoid sweeping jamming, significantly improving the communication success rate between the UAV and the user. Compared to other methods, our proposed algorithm achieves a higher average communication rate in intelligent jamming scenarios.

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